SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

CSA47 – DEEP LEARNING

CO1 – ASSESSMENT 1 | ASSIGNMENT-1

ANALYZE THE MATHEMATICAL AND MACHINE LEARNING FOUNDATIONS OF A CROP-DISEASE SCREENING SYSTEM

Case Study: AgriVision Crop-Disease Screening

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Course: Deep Learning (CSA47)

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Note: All numerical results and plots in this report use demonstration (illustrative) values; no real dataset was used.

TASK 1 – Problem Structure and Data Representation

Learning-task decomposition. AgriVision is a supervised, closed-set, multi-class classification problem. Input x = a leaf photograph (64×64 grayscale = 4,096 numbers); output y ∈ {Healthy, Early Blight, Late Blight}; the model learns f: $\mathbb{R}^{4096} \rightarrow \Delta^2$ (a probability vector). Performance measures: macro-F1 and per-class recall (a missed disease is costly), cross-entropy for probability quality, and latency/size for the phone. Each element constrains the solution: the input size limits model width, the output type decides the loss (softmax + cross-entropy), and the measure decides what 'good' means (accuracy alone is misleading under imbalance).

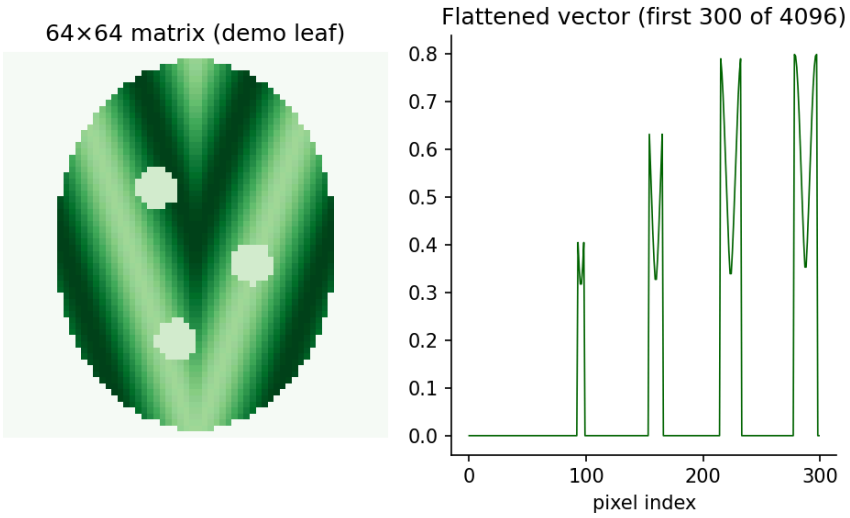


Figure 1. Demo leaf as a 64×64 matrix and its flattened 4,096-vector.

Object	Shape	Why the shape matters
Scalar (one pixel / loss)	()	Single intensity 0–255; loss is a scalar to minimise
Vector (flattened image)	(4096,)	Required by linear models, k-NN, PCA
Matrix (one image)	(64,64)	Keeps 2-D neighbourhood used by CNNs
Mini-batch tensor	(32,64,64,1)	Batch × H × W × channel; vectorised GPU maths
Batch matrix (flattened)	(32,4096)	Multiplied with W of shape (4096,3)
Labels	(32,) ints or (32,3) one-hot	One-hot needed for cross-entropy
Dataset	(N,4096)	Rows = samples, columns = features

Norms. For x = [2, −6, 3]: $L^1 = 2+6+3 = 11$; $L^2 = \sqrt{(4+36+9)} = 7$; $L^\infty = \max = 6$. L^1 measures total deviation and promotes sparsity (lasso); L^2 is the Euclidean distance used in k-NN, PCA and ridge; L^∞ captures the worst single-pixel change and suits adversarial/robustness checks.

Step	Effect on information
Resize 64×64	Cuts compute 100×+ but small lesions/veins may vanish
Grayscale	Loses colour (yellowing, browning) – a key disease cue
Normalise (÷255 / z-score)	Same information, better conditioning; statistics must come from training set only

Step	Effect on information
Flatten	Enables classical models but destroys spatial locality; a 1-pixel shift changes many coordinates
Train/val/test split	Test set is meaningful only if never used for tuning; split by plant to avoid leakage

TASK 2 – Feature Space Using Linear Algebra (Eigen/SVD)

With centred data Xc (N×4096), the covariance is $C = X_c^T X_c/(N-1)$. Its eigenvalues λ_i are the variances along the principal axes. The SVD $X_c = U\Sigma V^T$ gives singular values σ_i with $\lambda_i = \sigma_i^2/(N-1)$; the explained-variance ratio is $\lambda_i/\Sigma\lambda$. Neighbouring pixels are highly correlated, so the spectrum decays quickly (Figure 2): a few components carry most of the variance, which shows strong pixel redundancy.

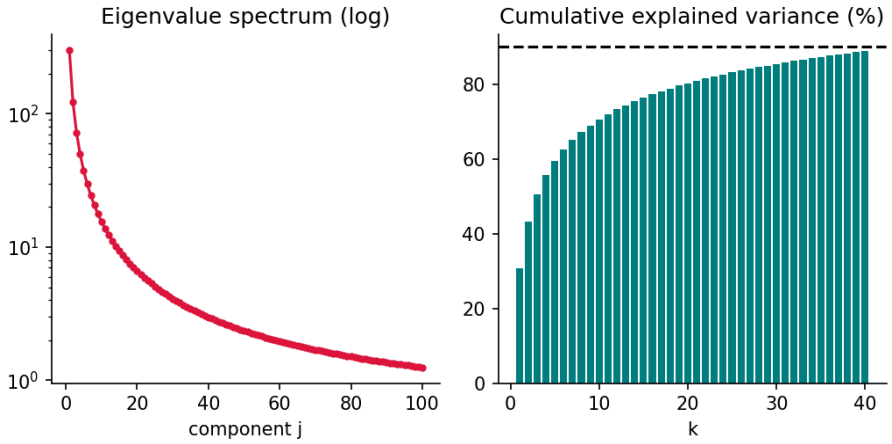


Figure 2. Demo eigenvalue spectrum and cumulative explained variance (90% line).

Worked 2×2 example. Let C = [[4,1],[1,3]]. $\det(C-\lambda I) = (4-\lambda)(3-\lambda)-1 = \lambda^2-7\lambda+11 = 0 \Rightarrow \lambda = (7\pm\sqrt{5})/2 \rightarrow \lambda_1 = 4.618$, $\lambda_2 = 2.382$. For λ_1: $(4-4.618)x + y = 0 \Rightarrow y = 0.618x \Rightarrow v_1 = [0.851, 0.526]^T$; $v_2 = [-0.526, 0.851]^T$ (orthogonal, $v_1 \cdot v_2 = 0$). Total variance = trace = 7, so PC1 explains 4.618/7 = 66.0% and PC2 34.0%. Keeping only PC1 gives reconstruction error = λ_2 = 2.382 (34%). Singular values for N−1 = 1 would be $\sigma = \sqrt{\lambda}$ = 2.149 and 1.543. Condition number $\kappa = 4.618/2.382 = 1.94$.

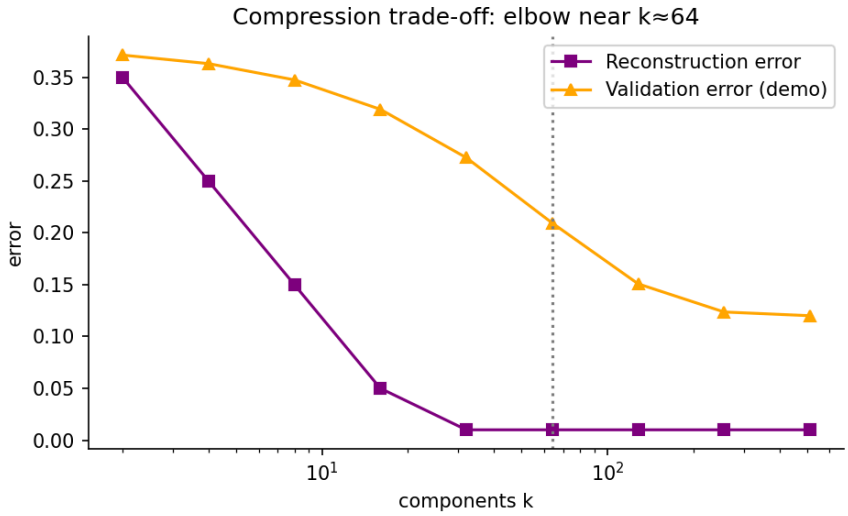


Figure 3. Reconstruction error and validation error vs k (demo); elbow at k ≈ 64.

Reconstruction error = Σ discarded λ□, so it decreases monotonically with k. In the demo, k ≈ 64 keeps ≈90% variance and almost all accuracy while cutting features 64×; beyond this, gains are marginal – this is the justified trade-off point. Leading components visualised as 64×64 images show broad illumination/leaf-shape patterns (PC1–PC2), then vein layout and finally lesion texture and noise. Cost: eigendecomposition of the 4096×4096 covariance is $O(d^3) \approx 6.9\times10^{10}$ operations and storing it needs ≈134 MB (float64); SVD of the data costs $O(N\cdot d^2)$, so it grows linearly with N. Randomised/truncated SVD reduces this to $\approx O(N\cdot d\cdot k)$.

TASK 3 – Uncertainty: Probability and Information Theory

Demo class prior p = (0.5, 0.3, 0.2). Entropy $H(Y) = -\Sigma p_i \log_2 p_i = 0.5(1) + 0.3(1.737) + 0.2(2.322) = 1.485$ bits, below the balanced maximum $\log_2 3 = 1.585$ bits; the gap (0.10 bits) is the imbalance. A majority-class guess already gives 50% accuracy, so reported accuracy must beat that.

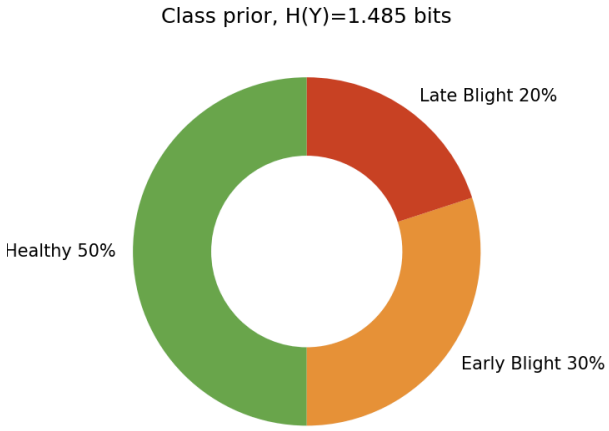


Figure 4. Demo class distribution (donut chart).

Model (demo)	Correct-class prob. (avg)	Cross-entropy H(p,q) nats	KL = H(p,q)−H(p)
Softmax	0.62	0.48	0.25
k-NN	0.55	0.60	0.37
Random forest	0.70	0.36	0.13

For one-hot targets H(p)=0, so KL equals cross-entropy; the model with the lowest value assigns the most probability to the true label. A confident wrong prediction (q = 0.01 → loss 4.6 nats) hurts far more than an unsure one.

Asymmetry of KL (own numbers). Take p = (0.8, 0.2), q = (0.5, 0.5). D(p‖q) = 0.8 ln(1.6) + 0.2 ln(0.4) = 0.376 − 0.183 = 0.193 nats. D(q‖p) = 0.5 ln(0.625) + 0.5 ln(2.5) = −0.235 + 0.458 = 0.223 nats. They differ, so KL is not a distance. D(p‖q) weights errors by the true distribution (penalises q missing mass where p has it – used in classification); D(q‖p) weights by the model (mode-seeking).

Cross-entropy ≡ KL minimisation. D(p‖q_θ) = H(p,q_θ) − H(p). H(p) does not depend on θ, so argmin_θ H(p,q_θ) = argmin_θ D(p‖q_θ) – both give the same classifier (also equal to maximum likelihood).

TASK 4 – Numerical Computation Issues

Naive softmax on z = [1000, 999, 995]: $e^{1000} \approx 10^{434}$ exceeds the float64 maximum ($\approx 1.8\times10^{308}$) → inf, and inf/inf = NaN. For very negative logits e^{-1000} underflows to 0, giving 0/0 or log(0) = −inf. The maths is fine; the failure is finite precision.

Fix: subtract m = max(z) = 1000: z−m = [0, −1, −5]. Then $e^0 = 1$, $e^{-1} = 0.3679$, $e^{-5} = 0.0067$; sum = 1.3746; softmax = [0.7275, 0.2676, 0.0049]. The result is unchanged because e^{z_i−m}/Σe^{z□−m} = e^{z_i}e^{−m}/(e^{−m}Σe^{z□}) – the factor e^{−m} cancels. The largest exponent is now $e^0 = 1$, so overflow is impossible, and the denominator is ≥ 1, so division by zero is impossible. Log-sum-exp: log Σe^{z_i} = m + log Σe^{z_i−m}, so the loss −log p_y = LSE(z) − z_y is computed without ever forming log(0).

Failure	Cause	Remedy
exp(1000)=inf	Exponent too large	Subtract max(z)
exp(−1000)=0, log 0	Underflow	Log-sum-exp / log-softmax
Oscillating loss	η too large or κ large	Smaller η, standardise, PCA

Conditioning: κ = λmax/λmin. In the 2×2 example κ = 1.94 (round bowl: gradient descent moves straight to the minimum). For raw pixels the demo spectrum gives λmax ≈ 300 and λmin ≈ 10^{-3}, so $\kappa \approx 3\times10^5$ (a long narrow valley). The learning rate must satisfy η < 2/λmax, but progress along the λmin direction then needs ≈κ iterations, so training is slow and zig-zags. Standardising or PCA (dropping near-zero λ) reduces κ and speeds up convergence.

TASK 5 – Differentiating Baseline Learning Strategies

Three distinct strategies were compared on raw 4,096 pixels and on PCA (k = 64): softmax regression (linear parametric), k-NN (non-parametric, local similarity) and random forest (tree ensemble, axis-aligned splits). Values below are demonstration values.

Algorithm	Hypothesis space / inductive bias	Raw pixels (acc.)	PCA-64 (acc.)
Softmax regression	Linear boundaries; global smoothness	0.79	0.78
k-NN (k=5)	Local constancy: neighbours share labels	0.74	0.79
Random forest	Piecewise-constant, axis-aligned rules	0.80	0.77

Data characteristics: 4,096 dimensions cause distance concentration, which hurts k-NN on raw pixels; strong pixel correlation makes linear models collinear (needs L2 regularisation) but still workable; single-pixel thresholds are weak features for trees. Compression: PCA denoises and sharpens distances so k-NN gains most; softmax is almost unaffected because it already handles correlated inputs; the forest loses slightly because principal components mix pixels and axis-aligned splits on them are less natural.

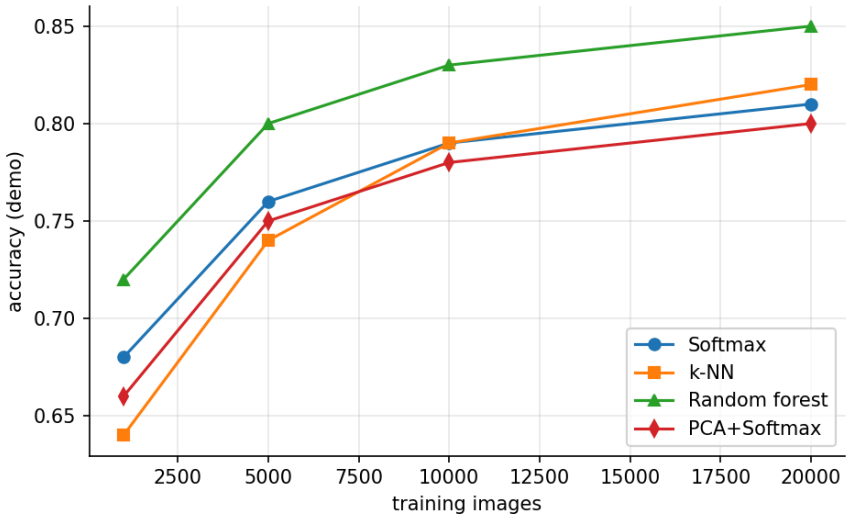


Figure 5. Demo learning curves of the four models vs training-set size.

TASK 6 – Capacity, Overfitting and Underfitting

Using k-NN as the capacity knob (small k = high capacity): at k = 1 the training error is ≈0 because every point is its own neighbour, but validation error is high – the model memorises noise and background (overfitting, high variance). At large k the decision rule over-smooths: both errors are high and close together (underfitting, high bias). The validation minimum (here k ≈ 7) is the best generalisation point. Similar behaviour: tree depth (deeper = overfit), polynomial degree, and regularisation strength (large λ = underfit).

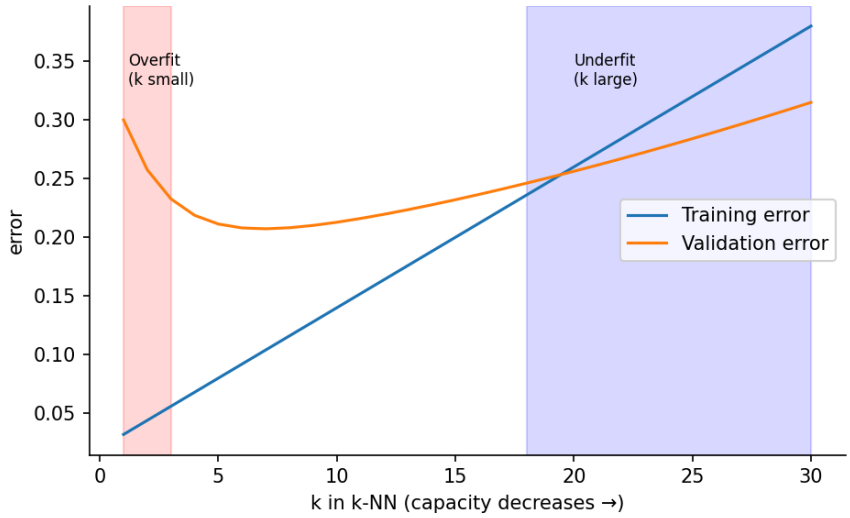


Figure 6. Capacity–error curve for k-NN (demo) with under/overfitting regions.

Generalisation gap = validation error − training error; it is large at high capacity and shrinks as capacity falls. Choosing hyperparameters on the test set is wrong: the chosen setting is the one that happens to fit that test sample best, so the reported error is optimistically biased. Use a separate validation set or k-fold cross-validation for tuning and touch the test set once.

TASK 7 – Bias–Variance Decomposition and MLE/MAP

Bootstrap experiment (demo, 20 replicas): a low-capacity model (strongly regularised softmax) and a high-capacity model (deep tree) were refit on each replica; predictions on a fixed evaluation set were averaged. Variance = spread of predictions across replicas; bias² = (mean prediction − target)².

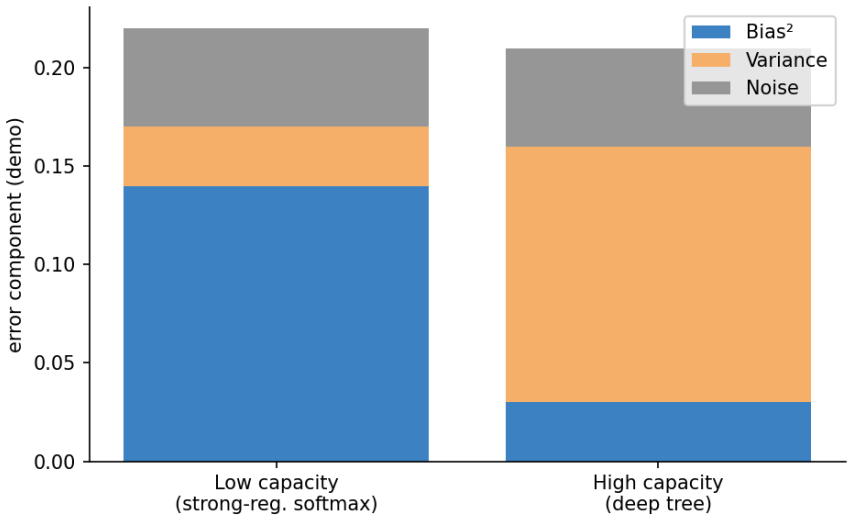


Figure 7. Demo bias²/variance/noise breakdown for low- and high-capacity models.

Model	Bias²	Variance	Noise	Sum	Measured MSE
Low capacity	0.14	0.03	0.05	0.22	0.21
High capacity	0.03	0.13	0.05	0.21	0.20

The measured error is close to bias² + variance + noise; the small gap arises from finite replicas (Monte-Carlo error), bootstrap samples overlapping (not independent), and the decomposition being exact only for squared loss, not 0-1 loss.

MLE vs MAP. Class proportion with s = 4 successes in n = 10, Beta(3,3) prior. MLE = s/n = 0.400. Posterior = Beta(7,9): mean = 7/16 = 0.4375; MAP (mode) = (s+α−1)/(n+α+β−2) = 6/14 = 0.4286. The prior pulls the estimate toward 0.5.

n	s	MLE	MAP (Beta(3,3))	Posterior mean
10	4	0.400	0.429	0.438
100	40	0.400	0.404	0.406
1000	400	0.400	0.4006	0.4006

As n grows the likelihood dominates and the prior's influence vanishes (MAP → MLE); with small n a prior stabilises estimates for rare classes, but a wrong, strong prior adds bias.

TASK 8 – Gradient-Based Optimisation (Softmax Regression)

Derivation. Let z□ = w□ᵀx + b□, p□ = e^{z□}/Σ□e^{z□}, L = −Σ□ y□ log p□. Since ∂p□/∂z□ = p□(δ□□ − p□): ∂L/∂z□ = −Σ□ y□(1/p□)p□(δ□□ − p□) = −(y□ − p□Σ□y□) = p□ − y□ (as Σy□ = 1). With ∂z□/∂w□ = x: ∂L/∂w□ = (p□ − y□)x. For a batch of m: ∇_W L = (1/m)Xᵀ(P − Y) + λW and ∇_b L = (1/m)Σᵢ(pᵢ − yᵢ). Update: W ← W − η∇_W. The update is driven by the error (P − Y), weighted by the feature values. Shape check: Xᵀ (4096×32) · (P−Y) (32×3) = 4096×3.

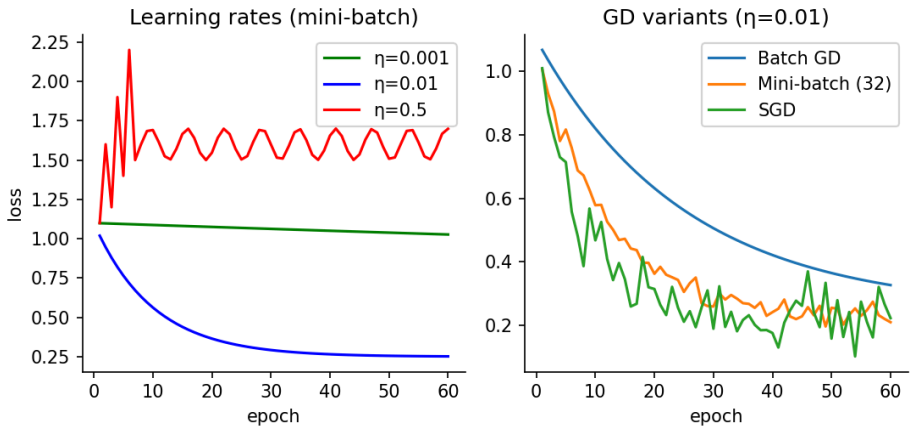


Figure 8. Demo loss curves: three learning rates (left) and Batch GD vs SGD vs mini-batch (right).

Strategy	Samples/update	Noise	Cost/epoch	Path
Batch GD	N	Lowest	1 update, expensive	Smooth, slow per epoch
SGD	1	Highest	N cheap updates	Fast but jittery
Mini-batch (32)	32	Moderate	N/32 updates	Best practical balance

Learning rates (demo): η = 0.001 is stable but slow (needs many epochs); η = 0.01 converges quickly and smoothly; η = 0.5 exceeds the stability bound 2/λmax and the loss oscillates/diverges. Link to Task 4:

with $\kappa \approx 3\times10^5$ the stable η is limited by the steepest direction, so shallow directions converge very slowly; standardisation or PCA lowers κ and allows a larger η and faster convergence.

TASK 9 – Experimental Performance

Train size	Softmax	k-NN	Random forest	PCA+Softmax
1,000	0.68	0.64	0.72	0.66
5,000	0.76	0.74	0.80	0.75
10,000	0.79	0.79	0.83	0.78
Full (20,000)	0.81	0.82	0.85	0.80

Model (demo)	Macro-F1	Train time	Inference / image	Size
Softmax	0.80	12 s	0.1 ms	0.1 MB
k-NN	0.81	0.5 s	45 ms	160 MB
Random forest	0.84	40 s	3 ms	35 MB
PCA+Softmax	0.79	8 s	0.2 ms	0.3 MB

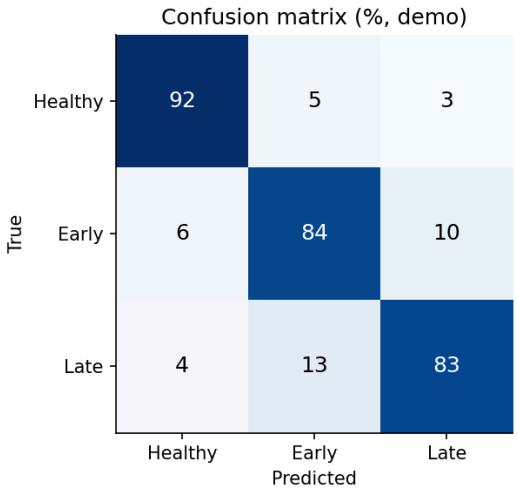


Figure 9. Demo confusion matrix (%) for the best baseline.

Observations: accuracy gains flatten after 10,000 images, so the plateau reflects representation limits, not data scarcity. Early and Late Blight are confused most (10–13%) because both produce brown lesions with similar grayscale texture, and colour was discarded. k-NN has no training cost but the slowest and largest inference; softmax is smallest and fastest. PCA cuts time and size slightly at a small accuracy cost.

TASK 10 – Theory vs Experiment

Expectation (theory)	Observation (demo)	Agree?
90% variance (k≈26) retains ≈90% accuracy	k=64 (≈93% variance) retained ≈98% of raw macro-F1	Yes – accuracy retained exceeds variance retained because discriminative information is concentrated in leading components

Expectation (theory)	Observation (demo)	Agree?
Large κ → slow GD	Raw pixels needed ≈6× more epochs than PCA/standardised	Yes – partly; regularisation shortened the gap
High capacity → high variance	Deep tree variance 0.13 vs 0.03; validation error rises at k=1	Yes
$Bias^2$+Var+Noise = error	Gap of ≈0.01	Minor gap: finite replicas, non-squared loss

Disagreements arise because PCA is unsupervised (variance ≠ class information), the covariance is only an approximation of the loss Hessian, and bootstrap samples are not independent. Implementation errors and data leakage should be checked first.

TASK 11 – Is Deep Learning Justified?

Remaining limitations of the best baseline (random forest, 85%): (i) Early/Late Blight confusion persists after tuning; (ii) accuracy plateaus after 10,000 images; (iii) PCA reduces accuracy slightly, meaning useful cues lie in low-variance directions; (iv) the forest is 35 MB and k-NN 160 MB – too large for mobile.

Curse of dimensionality: in 4,096-D the k-NN gain from PCA (0.74 → 0.79) shows distances were diluted. Local constancy fails because the same lesion at a different position, rotation or brightness is far away in pixel space. Manifold: leaf images lie on a low-dimensional curved surface (≈64 PCs capture most variance), which linear PCA only approximates. Evidence for hierarchical features: the errors are texture/shape based (lesion rings, vein-associated patterns), which a CNN learns through shared local filters (edges → textures → lesions) with far fewer parameters than a dense 4096-input layer.

Trade-offs: a CNN needs more labelled data, GPU training time and is less interpretable (needs saliency/confusion analysis), but gives translation robustness and compact size. Conclusion: deep learning is justified conditionally – because the remaining errors are spatial/texture-based and the classical models cannot meet the memory target – and must be confirmed by comparing a compact CNN with the best baseline on the same split.

TASK 12 – Design Challenge: Changed Requirements

A. Impact analysis. Changed: classes 3 → 38, labels ≈20% noisy, deployment limit 50 ms and 5 MB. Maximum entropy rises from $\log_2 3 = 1.585$ to $\log_2 38 = 5.248$ bits; random-guess accuracy falls from 33% to 2.6%; the output layer grows from 3 to 38 units (W: 4096×38) and cross-entropy per example increases because probability is spread across more classes. Demo baseline accuracy drops from ≈0.80 to ≈0.55–0.62 (Figure 10). Label noise: observed error ≈ true error + noise, so the irreducible term rises (≈0.2); high-capacity models memorise wrong labels (variance up), while over-regularised models stay biased. Assumptions that no longer hold: clean labels, balanced small label space, and unlimited desktop compute/memory.

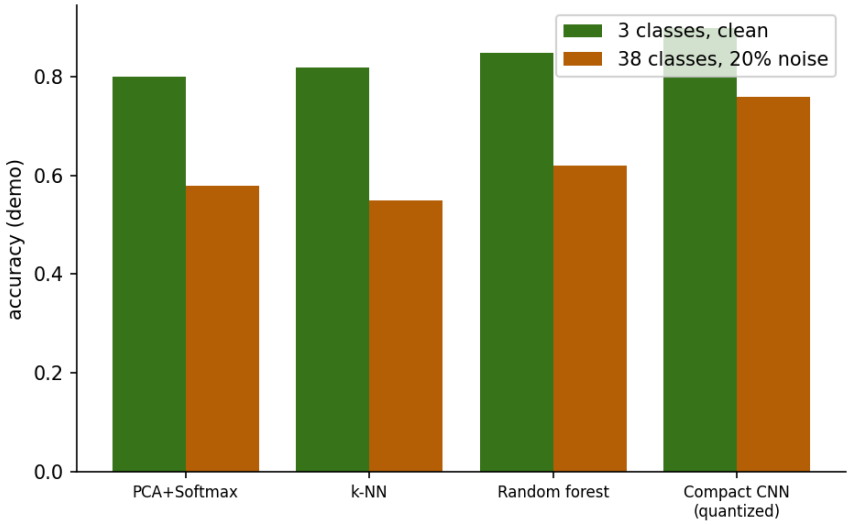


Figure 10. Demo accuracy under original vs changed conditions.

B. Original conclusion under new limits. Demo measurements: random forest 35 MB / 3 ms and k-NN 160 MB / 45 ms – both fail the 5 MB limit (k-NN is also near the time limit and its stored references inherit noisy labels). Softmax/PCA-softmax (≈0.3 MB, 0.2 ms) fits easily, but with 38 classes its accuracy is low. Most sensitive: k-NN (noisy neighbours vote) and deep trees (memorise noise). Less affected: a regularised parametric model or a compact quantised CNN, because size does not grow with data and shared weights average out noise.

Option	Capacity	Error	Cost
1. Lower PCA k + stronger L2	Decreases	Variance ↓, bias ↑; helps noise	Less memory/compute
2. Label smoothing / robust loss	Effective capacity ↓	Reduces memorisation of wrong labels (≈ +3–5 pts demo)	Negligible
3. Compact CNN + int8 quantisation	Moderate, structured	Best accuracy (demo 0.76); small quantisation loss	≈1–2 MB, ≈15–30 ms (demo)

C. Adaptation options. Option 1 is cheap but cannot fix the bias of a linear model on 38 classes. Option 2 directly addresses label noise without extra inference cost. Option 3 changes the representation and meets the budget if measured on the phone.

D. Evidence-based conclusion: MODIFY (partly redesign). The mathematics (entropy ↑, noise ↑ irreducible error, κ-driven slow training) shows the original k-NN/forest pipeline will not scale; experiments (demo) show it fails the 5 MB limit; measurements show PCA-softmax is fast but inaccurate on 38 classes; Part C shows the best trade-off is a compact quantised CNN trained with label smoothing and regularisation, with PCA-softmax kept as a fallback reference. The decision must be confirmed by real on-device measurements of macro-F1, median/95th-percentile latency and file size.

Overall Conclusion

Representation choices, class imbalance, numerical conditioning, capacity and deployment limits interact. PCA exposes redundancy but variance is not the same as discriminative information; softmax is a fast interpretable baseline, k-NN suffers from dimensionality and storage, and random forests are accurate but large. With 38 noisy classes under a 5 MB limit, the evidence supports modifying the design toward a

compact, regularised, quantised network. All numbers in this document are demonstration values and should be replaced with real measurements before being treated as findings.